Notice, question, predict

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use observations to make a prediction that can be tested.

SUCCESS CRITERIA - I can...

1. Separate stated evidence from unsupported claims.
2. Extend a position pattern and state an assumption.
3. Name an observation that could test the prediction.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “prediction” to explain your model or result. Sentence starters are optional.

evidence: observations or measurements that support an idea

prediction: what a model says may happen next

model: a drawing, explanation or equation for a physical situation

01 Read a scale

A ruler shows a toy car at 2 cm, then at 7 cm.
Draw both positions and the change. How far apart are the marks?

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Delta x = x_f - x_i$; $x_{next} = x_{now} + \Delta x$.
For a repeating position pattern only. i = initial; f = final. Check the pattern against evidence.

Worked reasoning: A cart is at 2, 5 and 8 m at 0, 1 and 2 s. Equal jumps of 3 m suggest 11 m at 3 s, if the pattern continues.

Your turn: Draw a position strip for these four times.
Mark which point is a prediction rather than an observation.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: ① ② ③

03 Notice a pattern

These are supplied cart observations, one second apart.
Write two things you notice.
Circle the part of the display that supports each notice.

Support: I notice ____. The data show ____.

Sentence starter: I notice ... . This supports ... because ... .

Time (s)	Position (m)
0	1
1	3
2	5

Marzano check - circle: ① ② ③

04 Make a prediction

Use the cart observations in 3.
Predict its position at 3 s.
Show how you extended the pattern.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Evidence or a claim?

Fictional briefing: “An object is approaching Earth. Its future path is uncertain.” Mark each E (stated evidence) or C (claim not established): approaching Earth; guaranteed impact; uncertain path.

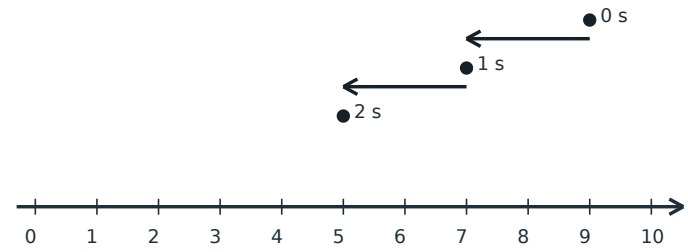
Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Test the model

A different cart is observed at 9, 7, 5 m at 0, 1, 2 s.
Predict its position at 3 s. What would a new reading of 4 m make you check?

Sentence starter: Both … . They differ in … because … .



Position (m); right is positive

Marzano check - circle: 1 2 3

07 Ask a useful question

Change “Is the asteroid bad?” into a question you could investigate.
Name one measurement you would need.

Support: I could investigate ____. I would measure ____.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 Choose useful evidence

Two groups time the same toy car over the same marked track. One records three times; the other writes "pretty fast." Which record would let someone check the claim? **Explain**.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

09 Exit check

A cart is at 0, 3, 6 m at 0, 1, 2 s.
Predict its position at 4 s. Include one assumption and one observation that could challenge your prediction.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

10 STRETCH: Apply it another way

Return to the worked example in problem 2. Create two possible positions at 4 s if the cart changes its motion after 2 s.
Draw both possibilities. What measurement would help choose between them?

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: 1 2 3